Reproducibility Report for Study by Salomons et al. (2026, Animal Behaviour)

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Introduction

[No abstract is needed.] Each reproducibility project will have a straightforward, no frills report of the study and reproducibility results. These reports will be publicly available as

supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of your attempt to reproduce the results. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for your reproducibility attempt, and any other essential information. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Justification for choice of study

The paper by Salomons et al. explains the cognitive development of assistance puppies (aged 8 to 20 weeks) as measured through performance on a biweekly battery of 9 tasks: marker gesture, working memory, working memory with distraction, auditory discrimination, odor discrimination, cylinder (inhibitory control and reversal learning), unsolvable, causal reasoning, and momentary pointing. The results support the multiple intelligences hypothesis and suggest the emergence of distinct cognitive abilities, with clear developmental breakpoints in marker gesture comprehension, inhibitory control, reversal learning, and eye contact occurring before 16 weeks, coinciding with the period of rapid brain development. The goal of this project is to answer if different cognitive skills develop at different rates in assistance puppies. The authors, however, chose to model each task separately and compare developmental rate visually rather than test comparisons directly. Using data exclusively from the 91 longitudinal subjects, I will reproduce Figure 2 and directly test whether age-related improvement differs across tasks using an age $\times$ task interaction. Next, I will simulate additional data from the fitted model to test whether a study of this size (N=91) can reliably detect these differences or is a larger sample (N=1000) is needed for sufficient power.

I am currently working to develop a battery of tests that can be administered to large numbers of dogs, in the hopes of identifying exceptional performers and determining whether strong performance on one task may generalize across different measures of the same ability and/or across cognitive domains. Knowing when different skills emerge will inform when puppies should be tested, and between-task correlations will suggest whether performance may generalize across domains. With the help of citizen science, this method may ultimately identify exceptional dogs.

Anticipated challenges

As explained above, my plan for this project includes (1) reanalyzing existing data from the 91 longitudinal puppies and (2) simulating additional subjects. My plan for this project will include 6 steps:

1. Reproduce Figure 2 with LOESS curve
2. Reproduce per-task models correct ~ age + (1 | puppy_ID) and correct ~ age + experience + (1 | puppy_ID)

3. Confirmatory mixed-effects logistic model, correct ~ age * task + (1 | puppy_ID) + (1 | puppy_ID:task)
    - Key result is age × task interaction (only including data from the 91 longitudinal subjects, so age and experience will confound)
    - Significant interaction supports multiple intelligences hypothesis
4. Calculate (for how much faster one skill develops relative to another)
5. Data simulation using fitted model (step 3) for N=91 vs N=1000
6. Plot simulated data with LOESS curves like Figure 2

My reproduction will be successful if curves match Figure 2, betas are within +/- 0.01 of the reported, and a find the same best-AIC model per task. My simulation will be successful if slopes show little bias, I can get ~95% CI coverage, and ~80% power for interaction, ideally at N=91.

I anticipate running into a few challenges when attempting to reproduce these results. First, age and experience are highly intertwined since they increase together for all longitudinal subjects. In addition, many of the dogs are siblings, half-siblings, cousins, etc. and this relatedness is not included in the dataset and may be a confound. Finally, of the 11 measures, 10 are binary responses (score out of 4 trials) and one is average looking time, thus I am considering removing the last measure from my simulation for ease. The biggest challenge, however, will be actually completing this project given my minimal programming knowledge and experience.

Learning Goals

- learn to fit and interpret mixed-effects models for the data
- learn how to write a simulation function that generates correlated data across all 11 measures for the same subject
- learn how to use simulation to test how sample size and testing schedule affect power (directly applies to planning my future project)

Links

Project repository (on Github): https://github.com/samkefer/salomons2026/tree/main

Original paper (as hosted in your repo): https://github.com/samkefer/salomons2026/blob/main/original_paper/original_paper

Methods

Description of the steps required to reproduce the results

1. **Data Preparation.** Before I could begin my reproduction, I first had to prepare the data. The dataset included in the supplements of Salomons et al. does not include subject ID and thus, you cannot see how an individual dog performs across tests. To fix this error, I used an additional spreadsheet (from a Shared Drive I can still access from my undergrad) to map subject names to each row and assign arbitrary subject ID numbers. After cleaning the xlxs file, each task was exported as a separate CSV file.
2. **Reproduce Figure 2.** Using the cleaned data, I filtered the each CSV to include only longitudinal subjects so I could reproduce Figure 2. For each of the 11 measures, I plotted proportion correct (or mean look time and proportion of trials looked for the unsolvable task) against age in weeks. Points were jittered for individual sessions, a LOESS smoothing curve was applied, and the 11 panels were combined into a single figure using the "patchwork" library.
3. **Reproduce Per-Task Models.** For each binary-outcome task, I fit two mixed-effects logistic regression models, with a random intercept for subject: correct ~ age + (1 | puppy ID) and correct ~ age + experience + (1 | puppy ID), where experience is the number of prior test sessions. The unsolvable task was modeled as proportion of trials looked (logistic) and mean look time (linear). Outcomes were modelled as the number of correct trials out of the total trials in each session. Models were run, and compared, on the longitudinal subjects alone and on entire 113-dog sample. For each sample, the two models were compared using AIC, and the age and experience coefficients of the longitudinal+control models were compared with those reported in the original paper.
4. **Confirmatory Age × Task Models.** The remainder of the analyses were completed using only binary-outcome tasks (i.e., excluding unsolvable task measures) and include only the longitudinal subjects (n=91). To test directly whether cognitive skills develop at different rates, I combined the nine binary-outcome tasks and fitted correct ~ age x task + (1 | puppy ID) + (1 | puppy ID:task) to the 91 longitudinal puppies, with age centred at 8 weeks. The key result is the age × task interaction, wherein a significant interaction indicates that tasks develop at different rates and supports the multiple intelligences hypothesis. Because age and experience are highly correlated in the longitudinal sample, this model cannot disentangle the effects of maturation and testing experience. I also examined the effects of adding a random puppy slope for age, refitting the model with all 113 puppies and task-specific experience effects to see if age can be separated from experience.
5. **Calculate Effect Sizes.** Using the confirmatory model, I extracted per-task effect size (age slope as an odds ratio) and compared each task. Comparing effect sizes serves as a measure of how much faster one skill develops relative to another. Slopes were also expressed as the change in proportion correct per week.
6. **Data Simulation.** Using both the fixed and random-effects confirmatory model, I will

simulate trial-level data for each task. Simulations will be run for two different samples: a replication of original sample size (n=91) and a larger sample (n=1000), whose testing schedules will be resampled from the original puppies.

7. **Plot Simulated Data.** Data from both the small and large simulations will be plotted in the same format as Figure 2 using jittered points and LOESS curves for each task. The simulated developmental trajectories can therefore be compared to the actual data and we can also observe the effects of increased sample size on the curves.

Differences from original study

To complete my analyses, participant ID had to be reconstructed by matching the provided summary data to raw and cleaned DCLB datasets, and a few sessions differed slightly between the three sources. Given the large scope of the original study, I needed to greatly limit the breatdth of my analysis for this project. Unlike the original study, my models only include age and experience, ignoring the potential effects of age and rearing strategy. With this narrowing, I will focus primairly on the longitudnial subjects and will thus not reproduce the longitudinal vs control comparisons of Table 3 or the segmented regression of Table 4. Additionally, the majority of my analyses were not included in the original paper, namely the combined age x task model, random slopes, the model including control puppies, and between-task effect sizes. Finally, Salomons et al. used RStudio 2023.06.1 while my project will use 2026.10.0.

Project Progress Check 1

Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

Results

Data preparation

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(readxl)
library(patchwork)
library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
auditory <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Auditory_Discr
```

```
Rows: 497 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
causal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Causal_Reasoning
```

```
Rows: 500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
detour <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Inhibit
```

```
Rows: 498 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
reversal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Revers
```

```
Rows: 496 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
marker <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Marker_Gesture.cs
```

```
Rows: 503 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
momentary <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Momentary_Poi
```

```
Rows: 495 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
odor <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Odor_Discriminatio
```

```
Rows: 492 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
unsolvable <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Unsolvable_Ey
```

```
Rows: 498 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (8): Dog ID, Age (wks), Age (days), Total Trials, Mean Look Time, # Tria...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
workingmem <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memo
```

```
Rows: 493 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
```

```
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
distraction <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Mem
```

```
Rows: 502 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Reproducibility

Individual Plots

```
auditory_long <- auditory |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

auditory_plot <- auditory_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
```

```
  ) +
  labs(title = "Auditory Discrimination")
```

```
causal_long <- causal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

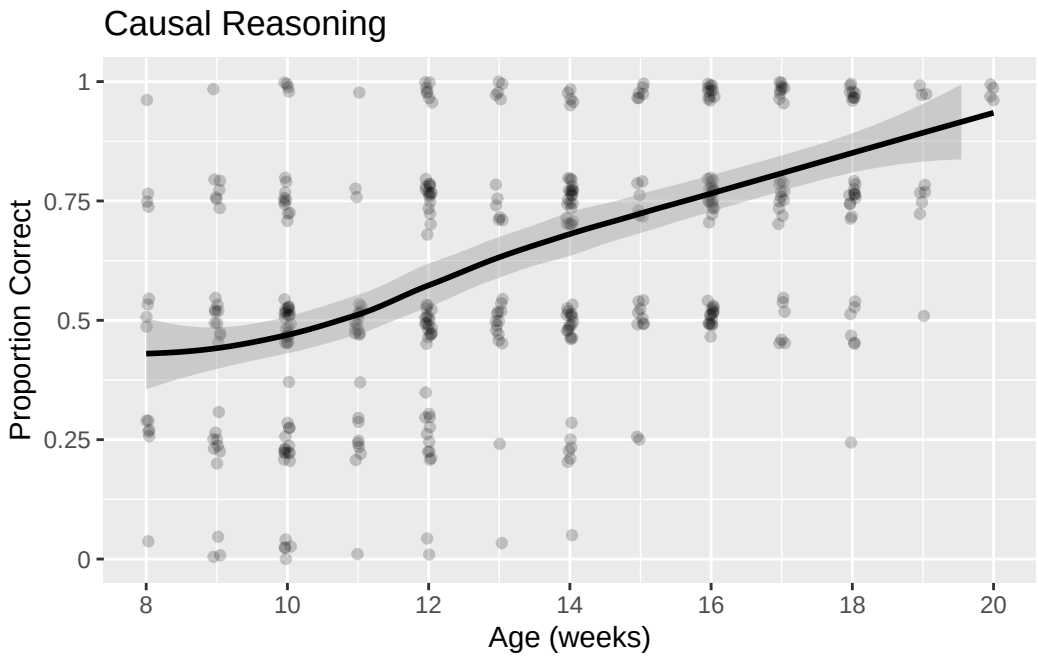
causal_plot <- causal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Causal Reasoning")

print(causal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 91 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
detour_long <- detour |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

detour_plot <- detour_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Cylinder: Inhibitory Control")
```

```r
distraction_long <- distraction |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

distraction_plot <- distraction_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2),
    labels = as.character
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Working Memory (Distraction)")
```

```r
marker_long <- marker |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

marker_plot <- marker_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
```

```
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Marker Gesture")
```

```
momentary_long <- momentary |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

momentary_plot <- momentary_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Momentary Pointing")
```

```
odor_long <- odor |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

odor_plot <- odor_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
```

```
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Odor Discrimination")
```

```
reversal_long <- reversal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

reversal_plot <- reversal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Cylinder: Reversal Learning")
```

```
unsolvable_long <- unsolvable |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Trials Looked` = as.numeric(`Proportion Trials Looked`),
```

```
    `Mean Look Time` = as.numeric(`Mean Look Time`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

proplooked_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Trials Looked`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Unsolvable (Proportion Looked)")

looktime_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Mean Look Time`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    name = "Mean Look Time (s)",
    limits = c(0, 20),
    breaks = seq(0, 20, by = 5)
  ) +
  labs(title = "Unsolvable (Looking Time)")
```

```
workingmem_long <- workingmem |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
```

```
  )

workingmem_plot <- workingmem_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name ="",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Working Memory (Delay)")
```

Figure 2

```
(marker_plot   | workingmem_plot | distraction_plot) /
(auditory_plot | odor_plot       | detour_plot)      /
(reversal_plot | looktime_plot   | proplooked_plot)  /
(causal_plot   | momentary_plot  | plot_spacer())&
  theme_grey(base_family = "serif") + #match theme to original
  theme(
    plot.title = element_text(hjust = 0.5, size = 11),
    axis.line  = element_line(colour = "black"),
    axis.title = element_text(size = 10),
    axis.text  = element_text(size = 9, colour = "black")
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 177 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 145 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 106 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 77 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 32 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 175 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 81 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 40 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 115 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

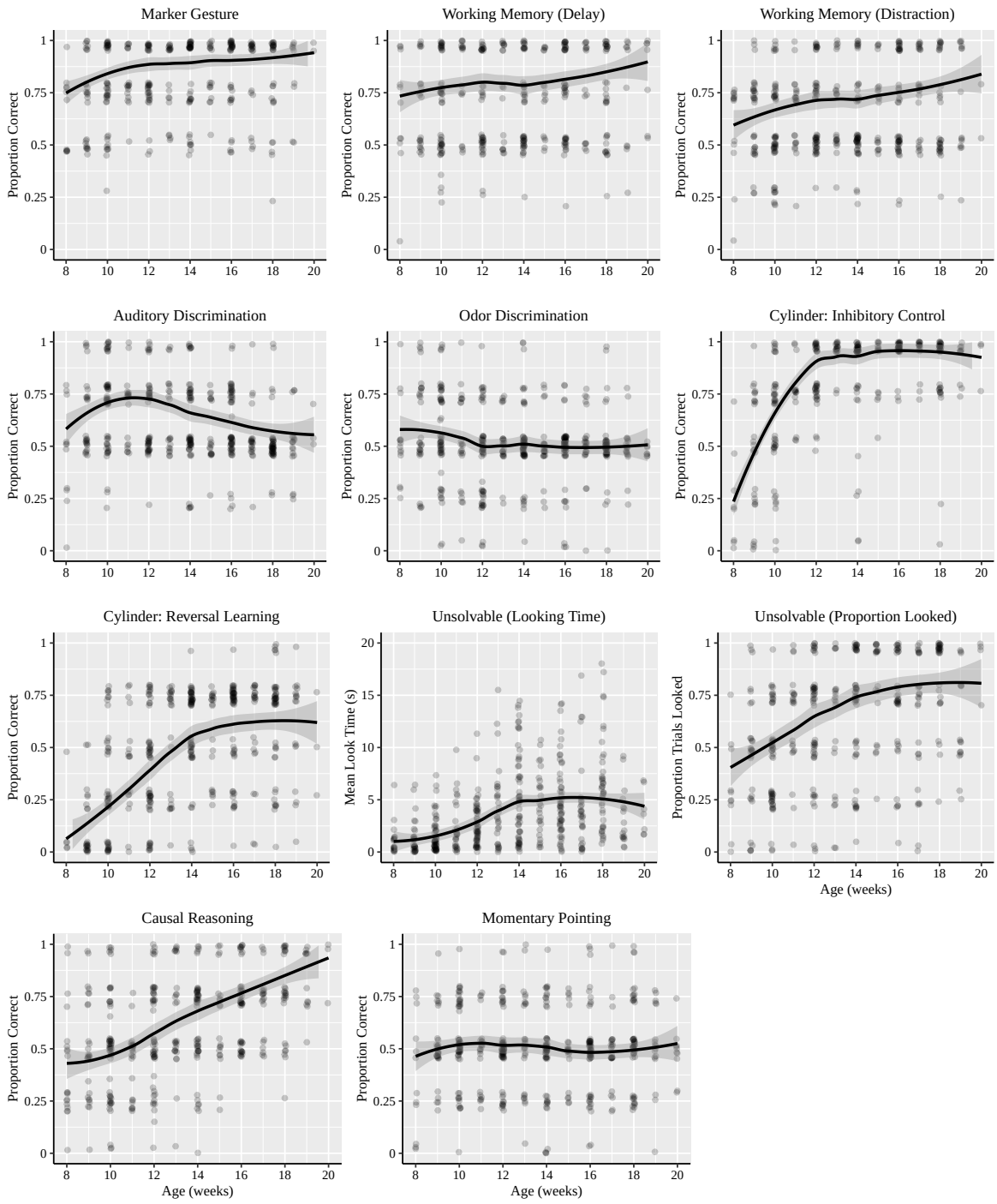
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 99 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 27 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Reproduce per-task models

Longitudinal only

```
task_models <- function(df) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                 family = binomial, data = df)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + exp
                 family = binomial, data = df)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble <- bind_rows(
  "Marker gesture"               = task_models(marker_long),
  "Working memory (Delay)"       = task_models(workingmem_long),
  "Working memory (Distraction)" = task_models(distraction_long),
  "Auditory discrimination"      = task_models(auditory_long),
  "Odor discrimination"         = task_models(odor_long),
  "Cylinder: inhibitory control" = task_models(detour_long),
  "Cylinder: reversal learning"  = task_models(reversal_long),
  "Causal reasoning"             = task_models(causal_long),
  "Momentary pointing"           = task_models(momentary_long),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   844.        846. age
2 Working memory (Delay)          1050.       1048. age + experience
3 Working memory (Distraction)    1168.       1169. age
4 Auditory discrimination         1226.       1221. age + experience
5 Odor discrimination             1167.       1169. age
6 Cylinder: inhibitory control     938.        895. age + experience
```

```
7 Cylinder: reversal learning    1312.       1298. age + experience
8 Causal reasoning               1210.       1186. age + experience
9 Momentary pointing             1219.       1219. age + experience
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable_long)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` + e
               family = binomial, data = unsolvable_long)

aic_tibble2 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble2
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1352.       1353. age
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + experience + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)


aic_tibble3 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble3
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2514.       2515. age
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 844.3)
- Working Memory with Distraction (AIC = 1168.1)
- Odor Discrimination (AIC = 1166.9)

- Unsolvable Proportion Looked (AIC = 1351.6)
- Unsolvable Average Looking Time (AIC = 2514.5).

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1047.9)
- Auditory Discrimination (AIC = 1220.9)
- Cylinder Inhibitory Control (AIC = 894.9)
- Cylinder Reversal Learning (AIC = 1297.8)
- Causal Reasoning (AIC = 1185.8).
- Momentary Pointing (AIC = 1218.8).

Longitudinal + Control, like paper

```
task_models2 <- function(df2) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                 family = binomial, data = df2)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + `#
                 family = binomial, data = df2)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble4 <- bind_rows(
  "Marker gesture"                = task_models2(marker),
  "Working memory (Delay)"        = task_models2(workingmem),
  "Working memory (Distraction)" = task_models2(distraction),
  "Auditory discrimination"       = task_models2(auditory),
  "Odor discrimination"          = task_models2(odor),
  "Cylinder: inhibitory control" = task_models2(detour),
  "Cylinder: reversal learning"  = task_models2(reversal),
  "Causal reasoning"              = task_models2(causal),
  "Momentary pointing"            = task_models2(momentary),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble4
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   876.        878. age
2 Working memory (Delay)          1081.       1073. age + experience
3 Working memory (Distraction)    1223.       1224. age
4 Auditory discrimination         1285.       1277. age + experience
5 Odor discrimination             1226.       1227. age
6 Cylinder: inhibitory control    1089.        957. age + experience
7 Cylinder: reversal learning     1427.       1364. age + experience
8 Causal reasoning                1327.       1244. age + experience
9 Momentary pointing              1269.       1270. age
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)

aic_tibble5 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble5
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1443.       1431. age + experience
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + `# Prior Test Sessions` + (1 | `Dog ID`),
              data = unsolvable, REML = FALSE)
```

```
aic_tibble6 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble6
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2640.       2629. age + experience
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 876.1)
- Working Memory with Distraction (AIC = 1223.1)
- Odor Discrimination (AIC = 1225.8)
- Momentary Pointing (AIC =)

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1072.8)
- Auditory Discrimination (AIC = 1276.8)
- Cylinder Inhibitory Control (AIC = 957.2)
- Cylinder Reversal Learning (AIC = 1364.4)
- Causal Reasoning (AIC = 1244.3).
- Unsolvable Proportion Looked (AIC = 1431.1)
- Unsolvable Average Looking Time (AIC = 2629.2).

Confirmatory mixed-effects logistic model

```
all_tasks <- bind_rows(
  "Marker gesture"               = marker_long,
  "Working memory (Delay)"       = workingmem_long,
  "Working memory (Distraction)" = distraction_long,
  "Auditory discrimination"      = auditory_long,
  "Odor discrimination"          = odor_long,
  "Cylinder: inhibitory control" = detour_long,
  "Cylinder: reversal learning"  = reversal_long,
```

```
  "Causal reasoning"             = causal_long,
  "Momentary pointing"           = momentary_long,
  .id = "task") |>
  transmute(task,
            id      = factor(`Dog ID`),
            age     = `Age (wks)` - 8,
            correct = `Trials Correct`,
            total   = `Total Trials`)

ctrl <- glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))

r_age_exp <- bind_rows(marker_long, workingmem_long, distraction_long, auditory_long,
                       odor_long, detour_long, reversal_long, causal_long,
                       momentary_long, unsolvable_long) |>
  summarise(r = cor(`Age (wks)`, `# Prior Test Sessions`)) |>
  pull(r)
```

```
# random intercepts only
m1 <- glmer(cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

# by-puppy random slope for age
m2 <- glmer(cbind(correct, total - correct) ~ age * task + (1 + age | id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

isSingular(m2)   # TRUE = random effects too complex for the data
```

```
[1] TRUE
```

```
anova(m1, m2)    # does the random slope improve fit?
```

```
Data: all_tasks
Models:
m1: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
m2: cbind(correct, total - correct) ~ age * task + (1 + age | id) + (1 | id:task)
   npar   AIC   BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
m1   20 10244 10371 -5102.0     10204
m2   22 10251 10391 -5103.4     10207     0  2          1
```

```r
# same slope as m2, no intercept-slope correlation
m3 <- glmer(cbind(correct, total - correct) ~ age * task + (1 + age || id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

isSingular(m3)    # FALSE = fitted cleanly
```

```
[1] FALSE
```

```r
anova(m1, m3)     # no sig improvement, m1 has slightly better AIC
```

```
Data: all_tasks
Models:
m1: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
m3: cbind(correct, total - correct) ~ age * task + ((1 | id) + (0 + age | id)) + (1 | id:tas
   npar   AIC   BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m1   20 10244 10371 -5102.0     10204
m3   21 10246 10379 -5101.8     10204 0.4436  1     0.5054
```

```r
# m3 most complex model that converged without warnings
m_final <- m3

# KEY RESULT: age x task interaction
m_final_noint <- update(m_final, . ~ . - age:task)
lrt_m3 <-anova(m_final_noint, m_final)

# m1 has marginally lower AIC so using as robustness check
m_final2 <- m1

# age x task interaction
m_final2_noint <- update(m_final2, . ~ . - age:task)
lrt_m1 <- anova(m_final2_noint, m_final2)
```

```r
all_tasks_all <- bind_rows(
  "Marker gesture"               = marker,
  "Working memory (Delay)"       = workingmem,
  "Working memory (Distraction)" = distraction,
  "Auditory discrimination"      = auditory,
  "Odor discrimination"          = odor,
  "Cylinder: inhibitory control" = detour,
  "Cylinder: reversal learning"  = reversal,
```

```
  "Causal reasoning"             = causal,
  "Momentary pointing"           = momentary,
  .id = "task") |>
  transmute(task,
            id         = factor(`Dog ID`),
            age        = `Age (wks)` - 8,
            experience = `# Prior Test Sessions`,
            correct    = `Trials Correct`,
            total      = `Total Trials`)

cor(all_tasks_all$age, all_tasks_all$experience)   #0.76
```

```
[1] 0.7659694
```

```
# Each task gets its own age slope AND its own experience slope
m_ae <- glmer(cbind(correct, total - correct) ~ age * task + experience * task +
                (1 | id) + (1 | id:task),
              family = binomial, data = all_tasks_all, control = ctrl)

# Does age x task matter AFTER accounting for task-specific experience effects?
m_ae_noint <- update(m_ae, . ~ . - age:task)
lrt_ae <- anova(m_ae_noint, m_ae)
lrt_ae
```

```
Data: all_tasks_all
Models:
m_ae_noint: cbind(correct, total - correct) ~ age + task + experience + (1 | id) + (1 | id:ta
m_ae: cbind(correct, total - correct) ~ age * task + experience * task + (1 | id) + (1 | id:t
           npar   AIC   BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
m_ae_noint   21 10658 10792 -5307.9     10616
m_ae         29 10630 10816 -5286.3     10572 43.33  8  7.613e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The age × task interaction was significant in the random-slope model (m3), χ^2(8) = 633.17, p <0.001, and in the random-intercepts model (m1), χ^2 (8) = 632.76, p <0.001. This indicates that the 9 tasks do not share one developmental rate, consistent with the multiple intelligences hypothesis. Because age and experience are highly correlated in the longitudinal puppies (r = 0.91), this model cannot disentangle the model whether these task differences reflect maturation, accumulated testing experience, or both.

After adding the 22 control puppies and task-specific experience effects, the age × task interaction was still significant, χ^2 (8) = 43.33, p <0.001. While part of the longitudinal age × task interaction effect may be shared with experience , the results from the 113-subject sample further support the multiple intelligences hypothesis, although age and experience remain partly correlated (r = 0.77).

Effect Sizes (Longitudinal)

```
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
# slopes on the probability scale
slopes <- emtrends(m_final, ~ task, var = "age")
slopes_table <- as_tibble(slopes) |>
  mutate(OR_per_week = exp(age.trend))
slopes_table
```

```
# A tibble: 9 x 7
  task                    age.trend     SE    df asymp.LCL asymp.UCL OR_per_week
  <fct>                       <dbl>  <dbl> <dbl>     <dbl>     <dbl>       <dbl>
1 Auditory discrimination  -0.0599  0.0151   Inf   -0.0896  -0.0302        0.942
2 Causal reasoning          0.231   0.0171   Inf    0.198    0.265         1.26
3 Cylinder: inhibitory c~   0.466   0.0292   Inf    0.409    0.523         1.59
4 Cylinder: reversal lea~   0.254   0.0167   Inf    0.222    0.287         1.29
5 Marker gesture            0.118   0.0227   Inf    0.0736   0.162         1.13
6 Momentary pointing       -0.00658 0.0144   Inf   -0.0348   0.0217        0.993
7 Odor discrimination      -0.0362  0.0145   Inf   -0.0647  -0.00777       0.964
8 Working memory (Delay)    0.0607  0.0186   Inf    0.0244   0.0971        1.06
9 Working memory (Distra~   0.0883  0.0164   Inf    0.0563   0.120         1.09
```

```
# age slope per task on the log-odds scale
slopes_prob <- emtrends(m_final, ~ task, var = "age", regrid = "response")
slopes_prob
```

```
 task                      age.trend      SE  df asymp.LCL asymp.UCL
 Auditory discrimination     -0.01343 0.00340 Inf  -0.02008  -0.00677
 Causal reasoning             0.05056 0.00373 Inf   0.04325   0.05787
```

```
 Cylinder: inhibitory control   0.04058 0.00281 Inf   0.03507   0.04608
 Cylinder: reversal learning    0.06237 0.00410 Inf   0.05433   0.07041
 Marker gesture                 0.01124 0.00210 Inf   0.00713   0.01534
 Momentary pointing            -0.00164 0.00360 Inf  -0.00870   0.00541
 Odor discrimination           -0.00904 0.00362 Inf  -0.01614  -0.00194
 Working memory (Delay)         0.00905 0.00276 Inf   0.00363   0.01446
 Working memory (Distraction)   0.01716 0.00316 Inf   0.01097   0.02335

Confidence level used: 0.95
```

```
prob_table <- as_tibble(slopes_prob) |>
  select(task, prob_per_week = age.trend)

effect_table <- as_tibble(slopes) |>
  transmute(task,
            OR      = exp(age.trend),
            OR_low  = exp(asymp.LCL),
            OR_high = exp(asymp.UCL)) |>
  left_join(prob_table, by = "task") |>
  arrange(desc(OR)) |>
  transmute(Task = task,
            `OR per week [95% CI]` = sprintf("%.2f [%.2f, %.2f]", OR, OR_low, OR_high),
            `Change in proportion correct per week` = sprintf("%+.3f", prob_per_week))
effect_table
```

```
# A tibble: 9 x 3
  Task                          `OR per week [95% CI]` Change in proportion cor~1
  <fct>                         <chr>                  <chr>
1 Cylinder: inhibitory control  1.59 [1.51, 1.69]      +0.041
2 Cylinder: reversal learning   1.29 [1.25, 1.33]      +0.062
3 Causal reasoning              1.26 [1.22, 1.30]      +0.051
4 Marker gesture                1.13 [1.08, 1.18]      +0.011
5 Working memory (Distraction)  1.09 [1.06, 1.13]      +0.017
6 Working memory (Delay)        1.06 [1.02, 1.10]      +0.009
7 Momentary pointing            0.99 [0.97, 1.02]      -0.002
8 Odor discrimination           0.96 [0.94, 0.99]      -0.009
9 Auditory discrimination       0.94 [0.91, 0.97]      -0.013
# i abbreviated name: 1: `Change in proportion correct per week`
```

```
# pairwise comparisons between tasks (Tukey-adjusted)
between_tasks <- as_tibble(pairs(slopes)) |>
```

```
  mutate(ratio_of_ORs = exp(estimate))
between_tasks
```

```
# A tibble: 36 x 7
   contrast                  estimate     SE    df z.ratio  p.value ratio_of_ORs
   <chr>                        <dbl>  <dbl> <dbl>   <dbl>    <dbl>        <dbl>
 1 Auditory discrimination ~  -0.291  0.0227   Inf -12.8   0                0.748
 2 Auditory discrimination ~  -0.526  0.0328   Inf -16.1   0                0.591
 3 Auditory discrimination ~  -0.314  0.0224   Inf -14.0   0                0.730
 4 Auditory discrimination ~  -0.178  0.0272   Inf  -6.55  2.05e- 9        0.837
 5 Auditory discrimination ~  -0.0533 0.0208   Inf  -2.57  2.00e- 1        0.948
 6 Auditory discrimination ~  -0.0237 0.0208   Inf  -1.13  9.69e- 1        0.977
 7 Auditory discrimination ~  -0.121  0.0238   Inf  -5.06  1.46e- 5        0.886
 8 Auditory discrimination ~  -0.148  0.0222   Inf  -6.69  8.11e-10        0.862
 9 Causal reasoning - Cylin~  -0.235  0.0337   Inf  -6.97  1.11e-10        0.791
10 Causal reasoning - Cylin~  -0.0231 0.0237   Inf  -0.976 9.88e- 1        0.977
# i 26 more rows
```

```
between_tasks_sig <- between_tasks |>
  mutate(sig = p.value<0.05) |>
  count(sig == "TRUE")
between_tasks_sig
```

```
# A tibble: 2 x 2
  `sig == "TRUE"`     n
  <lgl>           <int>
1 FALSE               8
2 TRUE               28
```

Each task's age slope is reported as an odds ratio (OR) per week. Effects were largest for cylinder inhibitory control (OR [95% CI] = 1.59 [1.51, 1.69]), cylinder reversal learning (OR = 1.29 [1.25, 1.33]), and causal reasoning (OR = 1.26 [1.22, 1.30]); smaller for marker gesture (OR = 1.13 [1.08, 1.18]), working memory with distraction (OR = 1.09 [1.06, 1.13]), and working memory with delay (OR = 1.06 [1.02, 1.10]); absent for momentary pointing (OR = 0.99 [0.97, 1.02]); and slightly negative for auditory discrimination (OR = 0.94 [0.91, 0.97]) and odor discrimination (OR = 0.96 [0.94, 0.99]).

When using a probability scale to look at proportion correct by age, cylinder inhibitory control increased by about 4 percentage points per week, less than cylinder reversal learning (about 6 points per week), likely because scores already neared ceiling by mid-testing. Comparing tasks directly showed that most developed at significantly different rates (significant for 28 of

36 comparisons), again reflecting the multiple intelligences hypothesis. For example, cylinder inhibitory control improved 1.65 times faster per week than odor discrimination ($p < .001$), whereas causal reasoning and reversal learning developed at similar rates (ratio = 0.98, $p = 0.99$).

Data Simulation

```
b     <- fixef(m1)
tasks <- levels(factor(all_tasks$task))

task_pars <- tibble(task = tasks) |>
  mutate(
    intercept = b["(Intercept)"] + coalesce(b[paste0("task", task)], 0),
    slope     = b["age"]         + coalesce(b[paste0("age:task", task)], 0)
  )
print(task_pars)
```

```
# A tibble: 9 x 3
  task                          intercept    slope
  <chr>                             <dbl>    <dbl>
1 Auditory discrimination          1.00   -0.0600
2 Causal reasoning                -0.560   0.231
3 Cylinder: inhibitory control    -0.380   0.466
4 Cylinder: reversal learning     -1.70    0.254
5 Marker gesture                   1.46    0.117
6 Momentary pointing               0.0319 -0.00654
7 Odor discrimination              0.305  -0.0362
8 Working memory (Delay)           1.16    0.0600
9 Working memory (Distraction)     0.530   0.0878
```

```
set.seed(2026)

vc        <- as.data.frame(VarCorr(m1))
sd_id     <- vc$sdcor[vc$grp == "id"]
sd_idtask <- vc$sdcor[vc$grp == "id:task"]
c(sd_id = sd_id, sd_idtask = sd_idtask)
```

```
    sd_id sd_idtask
0.1681470 0.4428252
```

```r
schedules <- all_tasks |> distinct(id, age) |> arrange(id, age)
real_ids  <- unique(schedules$id) #real testing schedules

simulate_m1 <- function(n_dogs, n_trials = 4) {

# Each simulated puppy: overall ability + a real puppy's testing ages
  dogs <- tibble(id       = seq_len(n_dogs),
                 u_id     = rnorm(n_dogs, 0, sd_id),
                 template = sample(real_ids, n_dogs, replace = TRUE))

# Task-specific ability: one value per puppy per task
  dog_task <- expand_grid(id = seq_len(n_dogs), task = tasks) |>
    mutate(v_idtask = rnorm(n(), 0, sd_idtask))

  dogs |>
    left_join(schedules, by = c("template" = "id"),
              relationship = "many-to-many") |>
    cross_join(task_pars) |>
    left_join(dog_task, by = c("id", "task")) |>
    mutate(
      p       = plogis(intercept + slope * age + u_id + v_idtask),
      total   = n_trials,
      correct = rbinom(n(), size = total, prob = p)   # 4 Bernoulli trials
    ) |>
    transmute(task,
              id          = factor(id),
              age,                                    # centered (weeks - 8)
              `Age (wks)` = age + 8,                  # for plotting
              total, correct,
              `Proportion Correct` = correct / total)
}

sim91   <- simulate_m1(91)
sim1000 <- simulate_m1(1000)
```

```r
m1_sim91   <- update(m1, data = sim91)
m1_sim1000 <- update(m1, data = sim1000)

m1_check <- tibble(term      = names(b),
                   true      = b,
                   sim_N91   = fixef(m1_sim91),
                   sim_N1000 = fixef(m1_sim1000))
```

```r
print(m1_check) #yay
```

```
# A tibble: 18 x 4
   term                                    true  sim_N91 sim_N1000
   <chr>                                  <dbl>    <dbl>     <dbl>
 1 (Intercept)                           1.00    1.00       1.01
 2 age                                  -0.0600 -0.0640    -0.0654
 3 taskCausal reasoning                 -1.56   -1.61      -1.62
 4 taskCylinder: inhibitory control     -1.38   -1.24      -1.38
 5 taskCylinder: reversal learning      -2.71   -2.68      -2.67
 6 taskMarker gesture                    0.462   0.703      0.467
 7 taskMomentary pointing               -0.971  -1.11      -0.961
 8 taskOdor discrimination              -0.697  -0.547     -0.678
 9 taskWorking memory (Delay)            0.160   0.112      0.142
10 taskWorking memory (Distraction)     -0.472  -0.518     -0.475
11 age:taskCausal reasoning              0.291   0.314      0.300
12 age:taskCylinder: inhibitory control  0.526   0.503      0.533
13 age:taskCylinder: reversal learning   0.314   0.321      0.315
14 age:taskMarker gesture                0.177   0.165      0.180
15 age:taskMomentary pointing            0.0534  0.0754     0.0560
16 age:taskOdor discrimination           0.0238  0.00991    0.0311
17 age:taskWorking memory (Delay)        0.120   0.125      0.123
18 age:taskWorking memory (Distraction)  0.148   0.164      0.152
```

```r
  (marker_plot   | workingmem_plot | distraction_plot) /
  (auditory_plot | odor_plot       | detour_plot)      /
  (reversal_plot | causal_plot     | momentary_plot)  +
  plot_annotation(
    title = "Original Data",
    theme = theme(
      plot.title = element_text(size = 20, hjust = 0.5)
    )
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 156 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 141 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 102 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 70 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 44 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 185 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 88 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

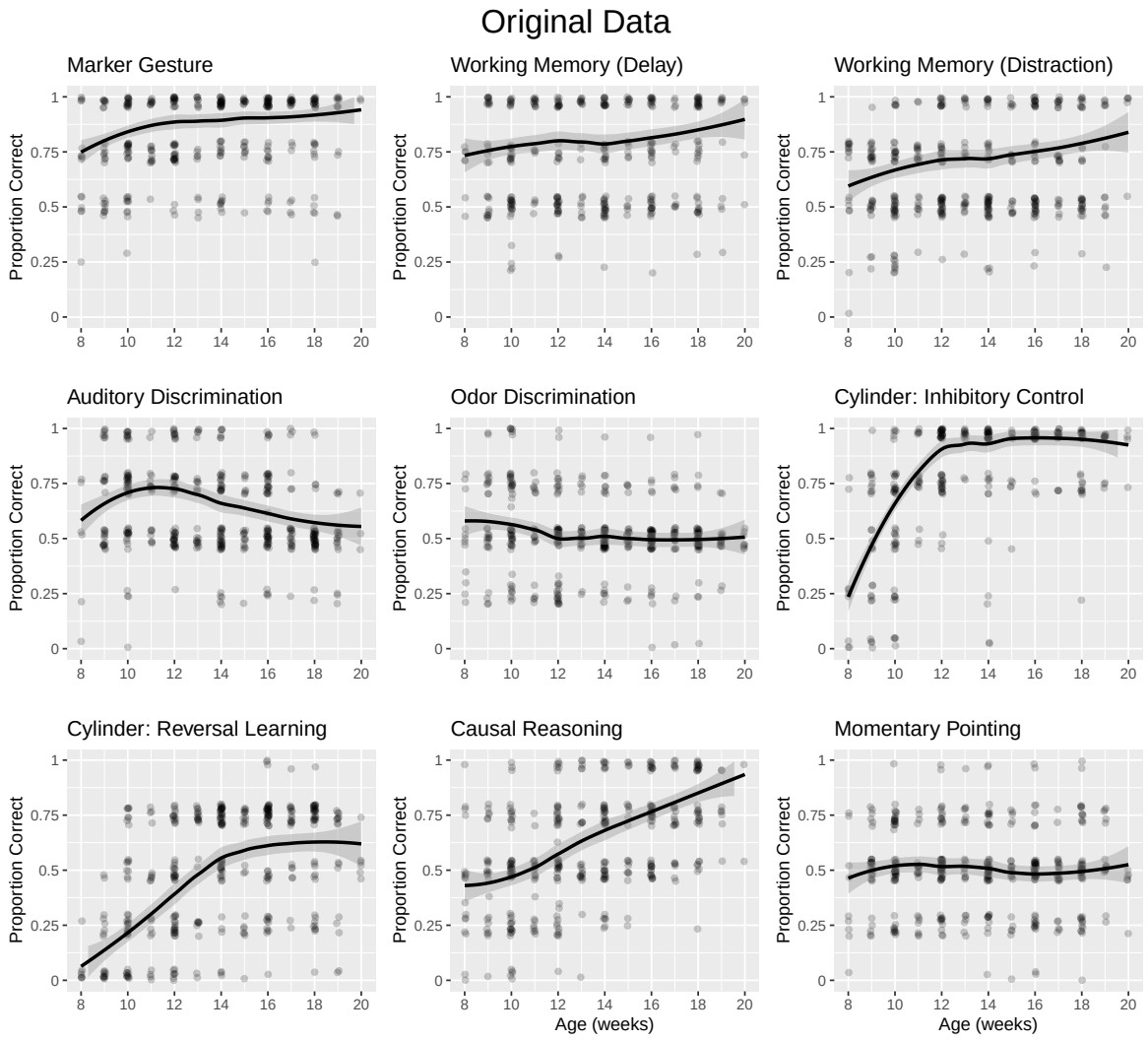
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 97 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 42 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
plot_task <- function(df, title, x_name = "", alpha = 0.2) {
  df |>
    ggplot(aes(x = `Age (wks)`, y = `Proportion Correct`)) +
    geom_jitter(alpha = alpha, width = 0.05, height = 0.05) +
    geom_smooth(method = "loess", color = "black") +
    scale_x_continuous(
      name = x_name,
      limits = c(8, 20),
      breaks = seq(8, 20, by = 2)
    ) +
    scale_y_continuous(
      limits = c(0, 1),
```

```
      breaks = seq(0, 1, by = 0.25),
      labels = as.character
    ) +
    labs(title = title)
}

sim_task <- function(task_name, title, x_name = "") {
  sim1000 |>
    filter(task == task_name) |>
    plot_task(title = title, x_name = x_name, alpha = 0.02)
}

marker_sim      <- sim_task("Marker gesture",               "Marker Gesture")
workingmem_sim  <- sim_task("Working memory (Delay)",      "Working Memory (Delay)")
distraction_sim <- sim_task("Working memory (Distraction)", "Working Memory (Distraction)")
auditory_sim    <- sim_task("Auditory discrimination",     "Auditory Discrimination")
odor_sim        <- sim_task("Odor discrimination",         "Odor Discrimination")
detour_sim      <- sim_task("Cylinder: inhibitory control", "Cylinder: Inhibitory Control")
reversal_sim    <- sim_task("Cylinder: reversal learning",  "Cylinder: Reversal Learning", "A
causal_sim      <- sim_task("Causal reasoning",             "Causal Reasoning",            "A
momentary_sim   <- sim_task("Momentary pointing",           "Momentary Pointing",          "A

  (marker_sim   | workingmem_sim | distraction_sim) /
  (auditory_sim | odor_sim       | detour_sim)      /
  (reversal_sim | causal_sim     | momentary_sim)  +
  plot_annotation(
    title = "Simulated Data (N = 1000)",
    theme = theme(
      plot.title = element_text(size = 20, hjust = 0.5)
    )
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2082 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1572 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1190 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 933 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 725 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1994 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1011 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

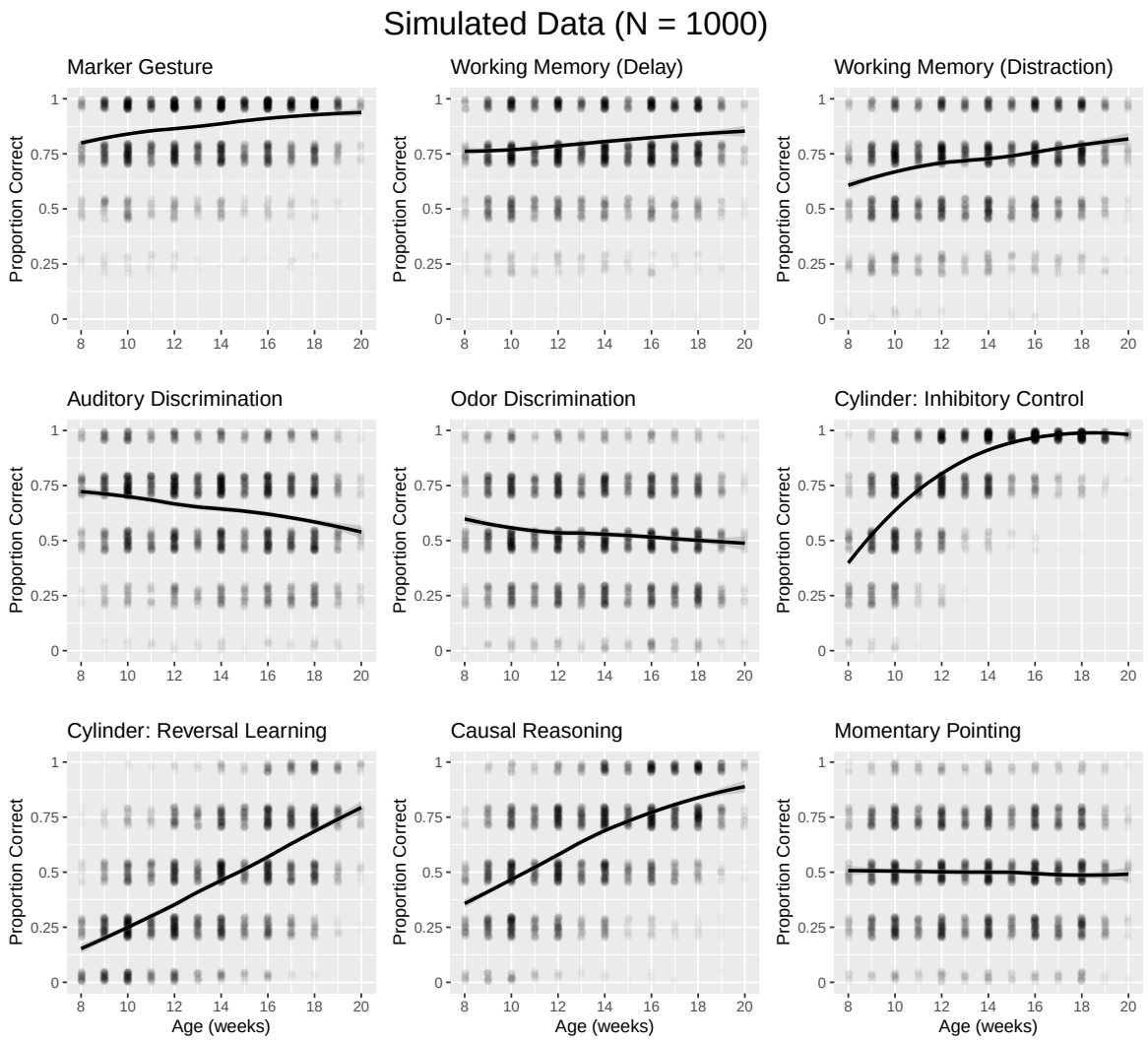
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1154 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 750 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
anova(update(m1_sim91,   . ~ . - age:task), m1_sim91) #sig agextask for n=91
```

```
Data: sim91
Models:
update(m1_sim91, . ~ . - age:task): cbind(correct, total - correct) ~ age + task + (1 | id) +
m1_sim91: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
                                 npar   AIC   BIC  logLik -2*log(L)  Chisq Df
update(m1_sim91, . ~ . - age:task)   12 13409 13488 -6692.6     13385
m1_sim91                             20 12639 12770 -6299.5     12599 786.15  8
                                 Pr(>Chisq)
update(m1_sim91, . ~ . - age:task)
```

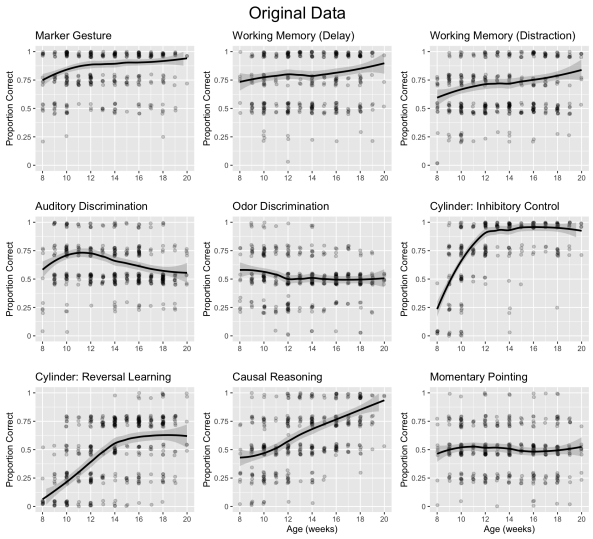
```
m1_sim91                              < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

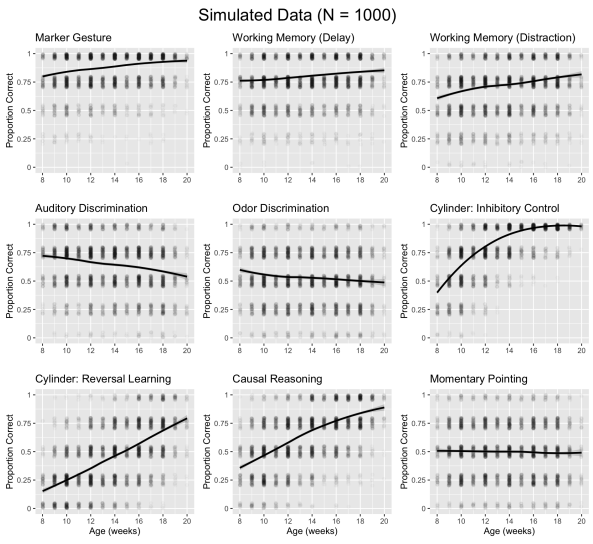
```
anova(update(m1_sim1000, . ~ . - age:task), m1_sim1000) #sig agextask for n=1000
```

```
Data: sim1000
Models:
update(m1_sim1000, . ~ . - age:task): cbind(correct, total - correct) ~ age + task + (1 | id)
m1_sim1000: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
                                     npar    AIC    BIC logLik -2*log(L) Chisq
update(m1_sim1000, . ~ . - age:task)   12 141803 141909 -70889    141779
m1_sim1000                             20 133446 133624 -66703    133406  8373
                                     Df Pr(>Chisq)
update(m1_sim1000, . ~ . - age:task)
m1_sim1000                            8  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Key analysis

The analyses as specified in the analysis plan.





Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.